

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 9: Global Navigation Satellite Systems (GNSS & GPS)

Syllabus Source: CAE Oxford (pp. 341-384) | Pass Target: $\geq 90\%$ AviationExam

1. THE NAVSTAR GPS ARCHITECTURE

GPS is a satellite-based radio navigation system operated by the US Space Force, structured into three segments:

- Space Segment: Baseline of 24 operational satellites distributed in 6 orbital planes inclined at 55° to the equator at an altitude of 20,200 km (semi-synchronous 12-hour orbit). Transmits on L1 (1575.42 MHz) and L2 (1227.60 MHz).
- Control Segment: Master Control Station (Schriever AFB, Colorado) and global monitor stations tracking orbits, updating clock corrections, and uploading ephemeris data.
- User Segment: Airborne receiver measuring Time of Arrival (TOA) of pseudo-random noise (PRN) codes from each satellite.

2. THE 4 SATELLITES FIX & CLOCK BIAS

Because the aircraft receiver clock is an inexpensive quartz crystal (not an atomic clock), it contains an unknown time error (clock bias):

- Mathematical Requirement: There are 4 unknown variables to solve: Latitude (X), Longitude (Y), Altitude (Z), and Receiver Clock Error (Δt). Therefore, A MINIMUM OF 4 SATELLITES IS MANDATORY FOR A 3D POSITION FIX!
- Dilution of Precision (DOP): Measures the geometric strength of satellite distribution. Best geometry (Low PDOP < 3): One satellite directly overhead, and three satellites spaced 120° apart near the horizon. Poor geometry (High PDOP > 6): Satellites bunched close together in the sky.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): GNSS Trilateration & Four Satellites

Mathematical Trilateration: - 1 Satellite: User is on surface of a sphere of radius R1. - 2 Satellites: Two spheres intersect in a circle in space. - 3 Satellites: Third sphere intersects circle at two unique points in space. - 4 Satellites: Why a 4th? Aircraft clock is an inexpensive quartz oscillator with clock bias! The 4th satellite eliminates the receiver clock error (Δt), solving for latitude, longitude, altitude, and exact time!

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Satellite Ranging & Pseudo-Range Correction

SCENARIO (GNSS Clock Bias Principle Drill): A GPS receiver measures time of signal arrival from a satellite: - The raw measured transit time indicates a distance of 21,000,000 meters (Pseudo-range) - The receiver clock has an internal bias of +0.000010 seconds (10 microseconds fast) - Speed of light $c = 300,000,000$ m/s QUESTION: What is the distance error caused by this 10 microsecond clock bias, and what is the true corrected range? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Calculate the distance error created by the clock bias: - Formula: Distance Error = Speed of Light (c) x Clock Bias (Δt) - Distance Error = $300,000,000 \text{ m/s} \times 0.000010 \text{ s} = 3,000 \text{ meters (3.0 km)}$! Step 2: Understand why 4 satellites are needed: - A tiny clock error of only 10 microseconds causes a massive 3-kilometer position error! - The 4th satellite provides the algebraic equation that solves for this exact 10 microsecond bias, eliminating the 3,000 m error completely! Step 3: Calculate the True Range: - True Range = Measured Pseudo-range - Distance Error - True Range = $21,000,000 \text{ m} - 3,000 \text{ m} = 20,997,000 \text{ meters}$. FINAL ANSWER: Distance error = 3,000 meters. True range = 20,997 km. The 4th satellite eliminates receiver clock bias.

[!] EXAM TRAP (AVIATIONEXAM) - 3 Satellites vs 4 Satellites Trap

- 3 satellites provide a 2D FIX (Latitude and Longitude ONLY), assuming altitude is already known! - 4 satellites are REQUIRED FOR A FULL 3D FIX (Latitude, Longitude, Altitude, and Clock Bias)! - Never answer 3 satellites when the question asks for a full 3D fix.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the MINIMUM number of satellites required by an unaugmented airborne GPS receiver to compute**

- [A]: 3 satellites
- [B]: AT LEAST 4 SATELLITES (to resolve X, Y, Z coordinates and receiver clock bias)
- [C]: 5 satellites
- [D]: 24 satellites

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The 4 unknowns (latitude, longitude, altitude, and receiver clock offset) require at least 4 independent pseudo-range equations from 4 visible satellites.

Q2: In satellite navigation, what does a LOW value of Position Dilution of Precision (PDOP < 3) indicate?

- [A]: Weak satellite signals
- [B]: EXCELLENT SATELLITE GEOMETRY (satellites widely spaced across the sky), yielding high navigational accuracy
- [C]: Clock failure
- [D]: High ionospheric delay

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Lower DOP values represent stronger geometric triangulation. PDOP < 3 represents ideal geometry with wide angular separation.

Q3: What is the primary source of pseudo-range ERROR in standard unaugmented single-frequency GPS?

- [A]: Rain attenuation
- [B]: IONOSPHERIC DELAY (refraction of radio waves through charged free electrons in the ionosphere)
- [C]: Relativistic error
- [D]: Antenna drag

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Ionospheric delay introduces unmodeled propagation delays of 5 to 30 meters, representing the largest single pseudo-range error source.